

Robust Real-Time DSP for a Hybrid Fiber and Photonic-Assisted Sub-THz Wireless Communication

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Abstract We demonstrate a real-time photonic-assisted sub-THz link. Across 10 km SSMF and a 1.4 m wireless link, the coherent 16-QAM system achieves 6.83 Gbit/s. Our fast-acquisition DSP performs real-time, non-data-aided timing recovery, channel equalization, and phase recovery on an FPGA, operating under ± 100 ppm clock offset and intermitted data. © 2026 The Author(s)

Introduction

Ultra-high-data-rate wireless communications is a primary enabler for sixth-generation (6G) mobile networks, supporting autonomous systems, extended reality, and the massive internet of things [1, 2]. In mobile fronthaul scenarios, sub-THz wireless links (above 100 GHz) provide a high-capacity alternative to optical fiber, bypassing the deployment costs and logistical complexity of physical infrastructure [1]. While the IEEE 802.15.3d standard provides a dedicated spectrum for these applications, operating at such high frequencies introduces severe free-space path loss and hardware limitations, necessitating high-gain directional antennas and robust digital signal processing (DSP) to maintain link stability [3]. Early milestones established the feasibility of the sub-THz band, achieving record data rates ranging from 100 Gbit/s to over 300 Gbit/s [4–7]. However, the majority of these experiments rely on computationally intensive offline DSP. While offline methods allow for complex compensation of channel impairments, they do not address the stringent latency and throughput requirements of practical 6G deployments. Real-time implementations, by contrast, are constrained by the hardware throughput of field-programmable gate arrays (FPGAs) and application-specific integrated circuits (ASICs), which must process several gigasamples per second (GSPS) with minimal power consumption and logic depth. A critical challenge for real-time sub-THz communication, especially when compared to established mmWave systems, is the DSP architecture. At multi-GHz bandwidths, standard serial processing is no longer viable, requiring highly parallelized architectures that can maintain synchronization and equalization without sacrificing real-time performance. While real-time sub-THz links have been demonstrated using ASICs in optical pluggable modules (e.g., QSFP) [8, 9], these are primarily optimized for the static, slow-drifting environments commonly seen in optical fiber. Consequently, they lack the agility required

to handle the rapid channel fluctuations and sudden signal dropouts inherent in wireless THz channels. Because sub-THz links are highly sensitive to transient physical obstructions and shadowing, the digital receiver must not only compensate for linear distortions but also demonstrate rapid re-synchronization and robust, non-data-aided adaptation following signal loss. These tasks are significantly more complex to implement in hardware than in offline software environments. To address these limitations, we demonstrate a photonic-assisted sub-THz wireless link utilizing optoelectronic heterodyne downconversion and a fully integrated, custom FPGA-based real-time DSP chain. Unlike previous works that rely on simplified intensity modulation schemes [10] or non-photonic systems [11, 12], we implement a coherent architecture utilizing an electro-optic (EO) frequency comb to provide a stable reference for the wireless carrier. This approach allows us to focus hardware resources on a robust, parallelized DSP chain capable of microsecond-scale re-acquisition and real-time equalization for highly dynamic environments at symbol rates of 2 Gbaud and above. We present a complete transmission system integrating a 10 km standard single-mode fiber (SSMF) link for remote processing with a sub-THz wireless link. The core of our contribution is the real-time FPGA implementation, which features a low hardware complexity non-data-aided Gardner timing recovery loop designed for microsecond-scale locking, a parallelized constant modulus algorithm (CMA) equalizer, and Viterbi-Viterbi carrier phase recovery. By employing QPSK and 16-QAM formats, the system achieves net data rates of up to 6.83 Gbit/s, demonstrating a scalable architecture for low-latency, high-capacity 6G fronthaul.

Photonic Real-Time Sub-THz Wireless Link

This work demonstrates a real-time sub-THz wireless link integrated with fiber-based transmission for remote broadband communications. The transmitter employs a coherent optical architecture based on an electro-optical (EO) frequency comb.

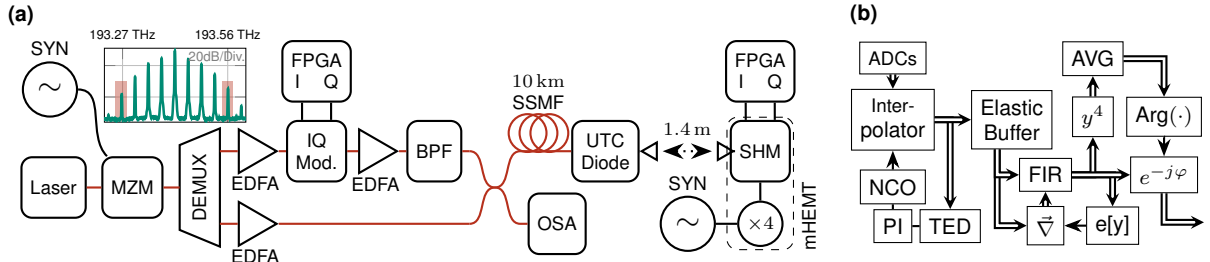


Fig. 1: (a) Experimental setup. A Mach-Zehnder modulator (MZM) generates a frequency comb, with lines selected via a wavelength de-multiplexer (DEMUX). The signal is modulated using an electro-optical IQ modulator and amplified by an erbium-doped fiber amplifier (EDFAs) and amplified spontaneous emission (ASE) noise is suppressed by a band-pass filter (BPF) before heterodyne downconversion in a uni-traveling carrier photodiode (UTC-PD). Wireless reception uses an electronic subharmonic mixer (SHM) based on a metamorphic high-electron-mobility transistor (mHEMT) architecture driven by a synthesizer (SYN) at 36.625 GHz. The link includes 10 km of standard single-mode fiber (SSMF). Inset shows the electro-optical (EO) comb spectrum obtained with an optical spectrum analyzer (OSA). (b) Real-time DSP architecture. The FPGA-based digital signal processing (DSP) chain processes data from two analog-to-digital converters (ADCs). A Gardner timing recovery loop utilizes a timing error detector (TED), proportional-integral (PI) filter, and numerically controlled oscillator (NCO). Adaptive equalization is performed by a finite impulse response (FIR) filter updated by constant modulus algorithm (CMA), followed by Viterbi-Viterbi carrier recovery for phase alignment.

Driven by a Mach-Zehnder modulator (MZM) and a synthesizer (SYN) at $f_{\text{FSR}} = 36.625$ GHz, the comb generates equally spaced lines used for both carrier and reference signals. As illustrated in Fig. 1(a), two comb lines with a 292 GHz separation are selected via two channels of a wavelength de-multiplexer (DEMUX). One line is IQ-modulated, amplified by an erbium-doped fiber amplifier (EDFA), and filtered to suppress amplified spontaneous emission (ASE) noise. This modulated signal is then combined with the unmodulated reference line in a 50/50 coupler. The resulting signal is converted to the sub-THz range using a uni-traveling carrier photodiode (UTC-PD), either directly or following transmission over a 10 km standard single-mode fiber (SSMF) link. The receiver utilizes a packaged WR-3.4 waveguide module based on Fraunhofer IAF's 35 nm mHEMT technology [13]. It consists of a three-stage low-noise amplifier (LNA) providing 28 dB gain and a 27 dB noise figure (220 – 330 GHz) followed by a second-harmonic sub-THz IQ mixer (SHM) driven by a 36.625 GHz local oscillator (LO) through a fourfold multiplication stage, all within the 35 nm mHEMT front-end package.

Real-Time Sub-THz DSP Implementation

The proposed system utilizes a Xilinx Zynq UltraScale+ RFSoc ZCU208 evaluation board for both baseband signal generation and real-time receiver DSP. To maintain a real-time throughput of 4.096 GSPS with an FPGA fabric clock frequency of $f_{\text{clk}} = 170.67$ MHz, the entire DSP chain is implemented using a parallel processing architecture with a factor of 24 samples per clock cycle. At the transmitter, a pseudo-random bit sequence (PRBS) with a length of $2^{11} - 1$ is generated and mapped to either QPSK or 16-QAM symbols. Pulse shaping is performed by a 17-tap root-raised cosine (RRC) filter with a roll-off factor of $\alpha = 0.3$. The resulting digital signals are converted to the analog domain by two integrated DACs operating at 4.096 GSPS with 12-bit resolution. The receiver

DSP is implemented with 8 bit fixed-point data paths, with higher precision maintained in feedback accumulators to ensure numerical stability. Timing synchronization is achieved via a non-data-aided feedback loop based on the Gardner algorithm [14]. The TED operates on the parallelized data stream, and the error signal is processed by a second-order PI loop filter. The estimated fractional delay is corrected using a 4-tap interpolator. To compensate for clock frequency offsets (CFO) between the transmitter and receiver, an elastic buffer with a depth of 52 samples is implemented, enabling reliable data transfer for both positive and negative clock offsets. As shown in Fig. 2, the loop achieves a stable lock within 10 – 30 μs , facilitating rapid re-acquisition after link interruptions especially caused by outages in the sub-THz wireless link. Post-synchronization, channel impairments

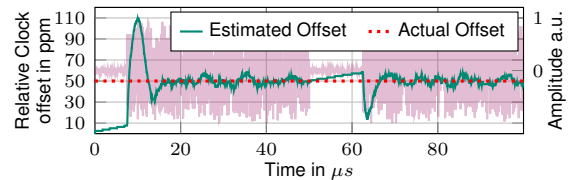


Fig. 2: Clock recovery locking for intermittent data transmission in VHDL simulation with -50 ppm clock offset.

and inter-symbol interference are mitigated using a 12-tap, $T/2$ -spaced adaptive equalizer. We employ the CMA [15] for blind adaptation. To reduce the hardware complexity of the weight update logic, a sign-error scheme is utilized, where filter coefficients are updated every clock cycle (f_{clk}). The equalizer simultaneously performs downsampling to one sample per symbol. Internal calculations for the tap updates utilize 16-bit precision to avoid quantization-induced instability. As illustrated in Fig. 3, the CMA equalizer is robust and maintains stability after initial convergence, which takes roughly 100 μs . Notably, since coefficient updates are proportional to the squared magnitude of the input signal, the algorithm is resistant to rapid divergence during outages, where the equalizer taps only marginally diverge. This makes the link recov-

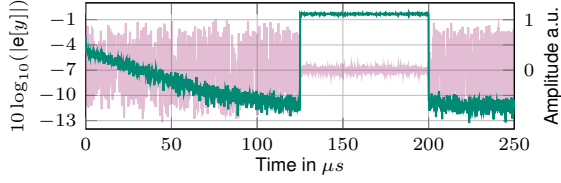


Fig. 3: CMA convergence with perfect clock synchronization in VHDL simulation. Initial convergence 100 μ s and re-convergence after outage of around 1 μ s.

ery time mostly depended on the clock recovery locking period, as shown in Fig. 2. To compensate for phase noise and residual frequency offsets from the intradyne conversion, a feed-forward carrier phase recovery stage is implemented using the Viterbi-Viterbi algorithm [16]. The y^4 operation is used to remove the modulation, and the phase is extracted using a LUT-based arctangent implementation. A moving average filter with a block size of 39 symbols is used to smoothen the phase estimate before phase rotation. The real-time FPGA implementation presented here is specifically tailored to the unique challenges of sub-THz wireless links. First, the high carrier frequencies in the sub-THz band are highly susceptible to atmospheric attenuation and physical obstructions, leading to frequent signal interruptions. Our implementation of a non-data-aided, Gardner-based timing recovery ensures that the link can re-synchronize within microseconds, providing the robustness required for reliable mobile backhaul or indoor applications. By offloading the DSP to a highly parallelized FPGA architecture, we demonstrate that processing high-bandwidth sub-THz signals can be achieved with performance comparable to fiber-based optical DSP, maintaining the low-latency performance necessary for 6G-and-beyond communication standards. Finally, the integration of adaptive equalization and carrier recovery directly into the hardware allows the system to mitigate the relatively high phase noise characteristic of high-frequency oscillators and optoelectronic heterodyne sources in real time.

Sub-THz Link Measurements

The real-time DSP is evaluated in a back-to-back (B2B) setup, directly connecting the DAC IQ outputs to the ADCs via a 1.2 GHz low-pass filter (LPF) and 3 dB attenuator. Constellation signal-to-noise and distortion ratio (SNDR) is measured over a ± 100 ppm clock offset range. Stable performance is observed, indicating reliable timing recovery, with mean SNDRs of 26.5 dB for QPSK and 26 dB for 16-QAM. The full link performance is evaluated using the setup shown in Fig. 1(a), employing the same LPF and 3 dB attenuators as in the B2B measurements. Two scenarios are considered. First without additional fiber, and second with 10 km of standard single-mode fiber (SSMF). Measurements are conducted for both QPSK and

16-QAM formats. The resulting system SNDR and bit-error ratio (BER) are presented in Fig. 4 for both scenarios. For QPSK, a stable SNDR of 14.9 dB (w/o SSMF) and 14.5 dB (w/ SSMF) is maintained across all clock offsets. Variations across both transmission cases and clock offsets remain below 0.3 dB, primarily attributed to minor experimental fluctuations such as polarization drift, which was more pronounced with the 10 km SSMF. All cases achieve BER below 10^{-7} , corresponding to a 4.096 Gbit/s line rate. For 16-QAM, the SNDR ranges from 13.8 dB (w/o SSMF) and 13.6 dB (w/ SSMF) with a BER below 2×10^{-2} for all relative clock offsets. This performance remains below the 20% overhead hard-decision forward error correction (HD-FEC) limit [17]. Consequently, a line rate of 8.192 Gbit/s is achieved, corresponding to a net data rate of 6.826 Gbit/s. The constellations for QPSK and 16-QAM are shown in Fig. 1.

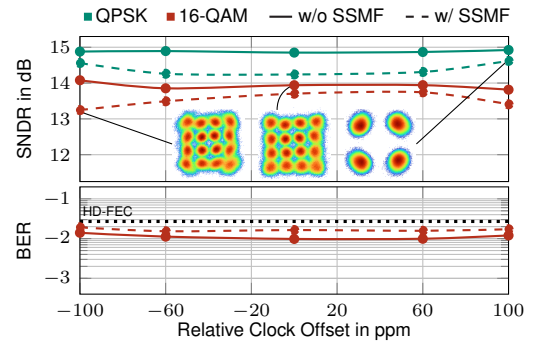


Fig. 4: SNDR and BER over clock frequency offset for the sub-THz wireless link with (w/) and without (w/o) 10 km SSMF. QPSK and 16-QAM constellations at 100 ppm clock offset.

Conclusions

This paper presents a real-time, photonic-assisted sub-THz link integrated with an SSMF transmission link. The system employs full real-time DSP on an FPGA to maintain sub-THz wireless connectivity despite intermittent transmission and clock frequency offsets of up to ± 100 ppm. VHDL fixed-point simulations of the feed-forward clock offset compensation architecture demonstrate rapid synchronization, achieving locking within 10 – 30 μ s. Analysis of the non-data-aided adaptive equalizer further confirms that no additional convergence penalties are introduced. By utilizing an electro-optical comb for reference tone generation, we achieved a stable sub-THz carrier while significantly relaxing phase noise and frequency offset requirements for the DSP. Experimental evaluations based on SNDR and BER indicate that the impact of relative clock offsets is negligible. This fully coherent architecture delivers a net data rate of 6.83 Gbit/s over a hybrid 10 km SSMF and 1.4 m wireless link, demonstrating a robust path toward seamless fiber-wireless integration in future 6G and sub-THz networks.

Acknowledgment

This work is financially supported by the German Federal Ministry of Research, Technology and Space (BMFTR) within the projects *Open6GHub* (16KISK010) and *Open6GHub+* (16KIS2405) and by the Dept. of Electronics and Telecommunications (DET) of Politecnico di Torino (*Incentivazione Dipartimentale – Azione A1.2*).

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